Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Assignment #4 - Heat Equation
Due Tuesday, October 20, 2026

1. (10 points) Let $k, L > 0$ and $f \in \mathrm{PC}^1[0, L]$ be a given function. Consider the heat equation

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{k}\frac{\partial u}{\partial t} = 0. \tag{HE}$$

(a) Write down the solution to (HE) with the boundary conditions

$$\frac{\partial u}{\partial x}(t, 0) = \frac{\partial u}{\partial x}(t, L) = 0, \qquad t > 0$$

and the initial condition

$$u(0, x) = f(x), \qquad 0 < x < L.$$

Solution — As with the wave equation, we apply an identical strategy of assuming the solution takes the separated form

$$u(t, x) = T(t)X(x)$$

Then

$$\frac{\partial^2 u}{\partial x^2} = T(t)X''(x) \qquad \frac{\partial u}{\partial t} = T'(t)X(x)$$

and plugging into the heat equation, we obtain:

$$k\frac{X''(x)}{X(x)} = \frac{T'(t)}{T(t)}$$

The left-hand side is a function of x only and the right-hand side is a function of t only. For them to be equal to one another for all values of t, x, they must both be equal to the same constant, which (for future convenience) we will notate as $-\lambda$. Hence our separated wave equation reduces to a system of independent, linear, homogeneous ordinary differential equations:

$$\begin{cases} kX''(x) + \lambda X(x) = 0 \\ T'(t) + \lambda T = 0 \end{cases}$$

We apply the characteristic equation to both ODEs, yielding solutions

$$A$$

For nontrivial $T(t)$, the boundary conditions reduce to

$$\frac{\partial u}{\partial x}(t,0) \implies T(t)X'(0) = 0 \implies X'(0) = 0$$

$$\frac{\partial u}{\partial x}(t,L) \implies T(t)X'(L) = 0 \implies X'(L) = 0$$

(b) Let $T_1 > 0$, take

$$f(x) = T_1 \frac{x}{L},$$

and determine the resulting solution by computing the Fourier coefficients and the corresponding Fourier series.

2. (15 points) Let $L, T_0 > 0$ and $f \in \mathrm{PC}^1[0, L]$ be given. Consider the temperature distribution of a long, thin bar in which heat is internally generated by a source with constant rate. In particular, we let $h > 0$ be given and consider the heat equation

$$\frac{1}{k}\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + h$$

for $0 < x < L$ and $t > 0$ with given boundary conditions

$$u(t,0) = T_0 \qquad \text{and} \qquad \frac{\partial u}{\partial x}(t,L) = 0$$

and initial temperature distribution

$$u(0,x) = f(x).$$

Determine the temperature throughout the bar at all later times, namely compute $u(t,x)$.

3. (13 points) Let c be a given number satisfying $\frac{\pi}{10} < c < \frac{9\pi}{10}$ and define for every $0 < x < \pi$ the function

$$h(x) = \begin{cases} 5, & \text{if } c - \frac{\pi}{10} < x < c + \frac{\pi}{10} \\ 0, & \text{else.} \end{cases}$$

(a) Solve the heat equation with $k = 1$, namely

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} = 0$$

with

$$u(t,0) = u(t,\pi) = 0, \qquad t > 0$$

and

$$u(0,x) = h(x), \qquad 0 < x < \pi.$$

Use the trigonometric identity

$$\cos(\alpha) - \cos(\beta) = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

to simplify the form of the Fourier coefficients.

(b) Notice that for t sufficiently large, we have

$$u(t,x) \approx B_1 e^{-t}\sin(x).$$

What choice of c as defined above makes B_1 largest?

(c) Using the idea in the previous part, answer the following question: Suppose that you're forced to put a cup of hot coffee somewhere between two blocks of ice. Where exactly should you put it if you want it to stay as warm as possible?

4. (12 points) Let $f, g \in \mathrm{PC}^1[0,\pi]$ be given and find the solution of the inhomogeneous wave equation with speed of propagation $c = 1$, namely

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + 1 = 0$$

for $0 < x < \pi$ and $t > 0$ subject to the given boundary conditions

$$u(t,0) = 0 \qquad \text{and} \qquad u(t,\pi) = 0$$

for $t > 0$ and initial conditions

$$u(0,x) = f(x) \qquad \text{and} \qquad \frac{\partial u}{\partial t}(0,x) = g(x)$$

for $0 < x < \pi$.